1. K-Nearest Neighbors (KNN)

1. A hospital has records containing patients' age, blood pressure, and glucose level. It wants to classify a new patient by looking at patients with similar measurements. Which approach best describes KNN?
A. Follow a series of learned if–else rules
B. Find similar patients and use their labels to classify the new patient
C. Fit one straight line through all patients
D. Build many decision trees

2. A KNN model for identifying plant species uses n_neighbors=5. A new plant's five nearest neighbors include three roses and two lilies. What will a standard KNN classifier predict?
A. Rose
B. Lily
C. Both species
D. It cannot make a prediction

3. A student builds KNN using patient age (18–90) and annual medical expenditure (0–500,000). The model seems to rely mostly on expenditure. What should the student consider?
A. Scaling the features before fitting KNN
B. Increasing the number of target classes
C. Removing the training labels
D. Changing the model to regression without checking the task

4. A team wants to predict the *price* of a house using prices of similar houses nearby. Which Python model fits this task?
A. KNeighborsClassifier
B. KNeighborsRegressor
C. GaussianNB
D. LogisticRegression

5. A researcher uses KNeighborsClassifier(n_neighbors=1) to classify noisy medical records. Training accuracy is very high, but test accuracy is poor. What is a reasonable explanation?
A. One unusual training example can strongly affect a prediction
B. KNN cannot classify medical records
C. KNN has stopped using distance
D. The model has too many decision trees

6. A student increases K from 3 to 200 in a small classification dataset. The model begins predicting the majority class for many different samples. Why?
A. A large neighborhood can hide local patterns
B. KNN automatically changes the target labels
C. Larger K always increases training accuracy
D. Feature scaling is no longer possible

7. A wildlife team has labeled measurements from 50,000 animals. Predictions with KNN are slower than expected. What is a likely reason?
A. Predicting may involve searching many stored examples for nearby animals
B. KNN retrains a random forest for every animal
C. KNN must draw a graph for every prediction
D. KNN cannot store more than 100 examples

8. A student wants closer patients to influence a KNN prediction more than patients farther away. Which setting is most relevant?
A. weights="distance"
B. weights="uniform"
C. max_depth=5
D. criterion="entropy"

9. A team is choosing between K values of 3, 5, 7, and 9. What is an appropriate way to select K?
A. Compare their performance using cross-validation on the training data
B. Choose the K that gives the highest score on the final test set repeatedly
C. Always choose the largest K
D. Choose K based only on the number of feature names

10. A student writes model = KNeighborsClassifier(n_neighbors=5) and then wants to train it. Which line should come next?
A. model.fit(X_train, y_train)
B. model.predict(y_train)
C. model.score(X_train)
D. model.transform(y_train)

2. Naive Bayes

1. An email service wants to classify messages as spam or legitimate using the words in each email. Why might Naive Bayes be a useful starting point?
A. It can use word-based evidence to compare the possible classes
B. It requires no labeled emails
C. It understands every sentence exactly as a human does
D. It measures the distance between email senders

2. A student converts emails into word-count features using CountVectorizer. Which model is a common choice for these features?
A. MultinomialNB
B. GaussianNB
C. KNeighborsRegressor
D. DecisionTreeRegressor

3. A medical dataset contains continuous measurements such as temperature, heart rate, and blood pressure. The student wants to try a Naive Bayes model that assumes a Gaussian distribution for each feature within each class. Which model should they choose?
A. GaussianNB
B. BernoulliNB
C. MultinomialNB
D. RandomForestRegressor

4. A news classifier represents each article by whether a particular word is present (1) or absent (0). Which Naive Bayes variant naturally matches this binary representation?
A. BernoulliNB
B. GaussianNB
C. KNeighborsClassifier
D. LinearRegression

5. In a training dataset, the word “prize” has never appeared in legitimate emails. Without smoothing, what problem can this cause?
A. Its estimated likelihood under the legitimate class may be zero
B. Every email will automatically become legitimate
C. The number of classes will double
D. The model will switch to KNN

6. A student sets MultinomialNB(alpha=1.0). What is the purpose of alpha in this situation?
A. Control additive smoothing
B. Set the number of neighboring emails
C. Set the maximum depth of a tree
D. Set the probability decision threshold

7. Two words, “chest” and “pain,” often appear together in clinical notes. What does the “naive” assumption simplify?
A. It treats feature evidence as conditionally independent given a class
B. It assumes all patients have the same diagnosis
C. It assumes each note contains one word
D. It removes all related words automatically

8. A dataset contains 90% legitimate emails and 10% spam. Before examining the words in a new email, which information can Naive Bayes use?
A. The class prior probabilities
B. The email’s true label
C. Its nearest neighbor’s distance
D. The depth of a decision tree

9. A student fits CountVectorizer on *all* emails before splitting into training and test sets. Why should the student revise this workflow?
A. The vocabulary preparation can use information from the test emails
B. Naive Bayes cannot work with word counts
C. The model requires every email to contain the same words
D. The training set must contain no text

10. A spam model achieves 96% accuracy, but nearly all spam messages are missed. Which result should the team inspect closely?
A. Recall for the spam class
B. The number of characters in the Python file
C. The number of feature names alone
D. The model’s training time alone

3. Decision Trees

1. A doctor wants a model that can produce rules such as “if glucose exceeds a threshold, then check age.” Which model naturally produces this style of decision process?
A. Decision tree
B. KNN
C. Naive Bayes
D. Linear regression

2. A bank trains a decision tree to classify loan applications. The first question in the tree is whether income exceeds ₹50,000 per month. What is this first question called?
A. A root-node split
B. A class probability
C. A nearest neighbor
D. A regression coefficient

3. A student’s tree correctly classifies almost every training example but performs poorly on new examples. The tree has many tiny leaves. What should the student consider?
A. Restricting tree depth or increasing min_samples_leaf
B. Increasing tree depth without checking validation results
C. Removing the test set
D. Changing all target labels to the majority class

4. A hospital wants to predict whether a patient will be readmitted: yes or no. Which Python model matches this target?
A. DecisionTreeClassifier
B. DecisionTreeRegressor
C. KNeighborsRegressor
D. LinearRegression

5. The same hospital now wants to predict the *number of days* a patient will remain admitted. Which tree model matches this target?
A. DecisionTreeRegressor
B. DecisionTreeClassifier
C. BernoulliNB
D. KNeighborsClassifier

6. A tree uses the rule temperature <= 38.0. A patient has a temperature of 37.5°C. What does the tree do?
A. Follow the branch where the condition is true
B. Follow both branches
C. Stop making predictions
D. Change the temperature to 38.0°C

7. A team needs to explain a small tree's predictions to clinicians. Which Python function can help display its learned splits?
A. plot_tree
B. train_test_split
C. StandardScaler
D. accuracy_score

8. A student sees that a feature has high feature_importances_ in a fitted tree. Which conclusion is justified?
A. The fitted tree relied strongly on that feature for its splits
B. The feature has been proven to cause the outcome
C. The feature will always be important in every dataset
D. The feature should be removed immediately

9. A tree for predicting disease at hospital admission includes a laboratory result obtained three days after admission. What is the main concern?
A. The feature would not be available when the prediction is needed
B. The laboratory result is numerical
C. A tree cannot use medical data
D. The tree has too few classes

10. A student wants to compare max_depth values of 2, 4, and 8. What is the best purpose of doing so?
A. Find a depth that balances useful patterns and generalization
B. Force all trees to make identical predictions
C. Determine the number of target classes
D. Avoid fitting the model

4. Random Forest

1. A single decision tree gives noticeably different predictions when the training data changes slightly. A team combines many trees trained with randomness. What model are they building?

A. Random forest
B. Logistic regression
C. Gaussian Naive Bayes
D. KNN regression

2. A researcher creates RandomForestClassifier(n_estimators=100). What does n_estimators=100 specify?

A. The forest contains 100 trees
B. The training set has 100 rows
C. The target has 100 classes
D. Each tree has depth 100

3. A forest classifies a transaction as fraudulent when most of its trees vote "fraud." What is this an example of?

A. Combining predictions from multiple trees
B. Finding the nearest transaction
C. Using a single linear equation
D. Removing the minority class

4. A property company wants to predict a house's numerical sale price using many trees. Which class should it use?

A. RandomForestRegressor
B. RandomForestClassifier
C. BernoulliNB
D. KNeighborsClassifier

5. A student asks why each tree in a forest does not receive exactly the same training rows. What is the purpose of bootstrap sampling?

A. To create variation by sampling rows with replacement
B. To copy the test set into training
C. To make every tree identical
D. To standardize every feature

6. A fraud-detection dataset has 99% legitimate transactions. A forest gets 99% accuracy but detects almost no fraud. What should the team examine?

A. Precision and recall for the fraud class
B. Only the total number of trees
C. Only the accuracy score
D. Only the names of the features

7. A fitted forest reports high importance for "customer age." What is a careful interpretation?

A. The forest used age substantially, but the result does not prove age causes the outcome
B. Age has been proven to cause the outcome
C. Every future dataset will give age the same importance
D. Age is necessarily the only feature needed

8. A researcher sets oob_score=True while using bootstrap sampling. Which samples contribute to the out-of-bag estimate for a particular tree?

A**. Training samples not selected for that tree's bootstrap sample**
B. The final test samples
C. Only samples with missing labels
D. Every sample used to fit that tree

9. A team increases n_estimators from 50 to 500. Which trade-off should they expect?
A. More computation, without a guarantee of better test performance
B. A guaranteed increase in test accuracy
C. Automatic conversion into a single tree
D. Removal of all class imbalance

10. A hospital forest uses a feature recorded after a patient is discharged to predict risk at admission. The validation score is excellent. What should the team do?
A. Remove the future-information feature and reevaluate
B. Deploy immediately because the score is high
C. Add more copies of that feature
D. Increase the number of trees until the problem disappears

5. Logistic Regression

1. A clinic wants to predict whether a patient will develop a condition within one year: yes or no. Which model is a suitable simple classification baseline?
A. LogisticRegression
B. LinearRegression for an unrestricted numerical output
C. RandomForestRegressor
D. KNeighborsRegressor

2. A fitted logistic regression model gives a patient a positive-class probability of 0.78. The clinic uses a threshold of 0.5. What class will the system assign?
A**. Positive**
B. Negative
C. Both classes
D. No class

3. A screening program is especially concerned about missing patients who have a disease. What might it consider doing to its positive decision threshold?
A. Lowering it, while evaluating the resulting false alarms
B. Raising it to 1.0 without evaluation
C. Removing the threshold entirely
D. Setting every feature to zero

4. A student lowers the positive-class threshold from 0.5 to 0.2. What is a possible result?
A. More actual positive cases are identified, along with more false alarms
B. All false positives disappear
C. The model becomes a decision tree
D. The fitted coefficients are automatically retrained

5. A model uses age in years and income in rupees. The student wants regularization and optimization to behave sensibly across features with very different scales. What should they consider?
A. Standardizing the input features
B. Removing the target labels
C. Setting n_estimators=100
D. Choosing an odd K

6. A student writes LogisticRegression(C=0.01) and compares it with LogisticRegression(C=100). Which setting generally applies stronger regularization?
A**. C=0.01**
B. C=100
C. Both always apply identical regularization
D. Neither model uses regularization

7. A fitted model has a positive coefficient for blood pressure when predicting a condition. What does that coefficient suggest, with the other supplied features held fixed?
A. Higher blood pressure increases the model's log-odds of the positive class
B. Blood pressure is proven to cause the condition
C. Higher blood pressure must reduce the predicted probability
D. The model ignores blood pressure

8. A disease occurs in only 1% of patients. A model predicts "no disease" for everyone and achieves 99% accuracy. What should the student check?
A. Recall, precision, and the confusion matrix for the disease class
B. Only overall accuracy
C. The number of Python imports
D. Only the number of features

9. While training, scikit-learn reports that logistic regression did not converge. What is a sensible next step?
A. Check feature scaling and consider solver settings or max_iter
B. Assume the model is perfectly trained
C. Set n_neighbors=5
D. Increase the number of trees

10. A hospital model predicts disease at admission but uses the final discharge diagnosis as an input. Why is its impressive test result questionable?
A. The input contains information unavailable at admission
B. Logistic regression cannot use categorical targets
C. The model needs at least 100 features
D. Predicted probabilities can never be evaluated